

# ADAM TANG

+1 (301) 275-6996 | US Citizen | [adamtang0715@gmail.com](mailto:adamtang0715@gmail.com) | [LinkedIn](#) | [Github](#) | [Portfolio Website](#)

## EDUCATION

**University of California, Berkeley** | GPA: 4.0 (so far) Berkeley, CA  
*Bachelor's in Electrical Engineering and Computer Sciences, Minor in Environmental Engineering Science* *Expected June 2029*

- **Relevant Coursework:** CS61A - Python, Scheme, SQL | Math 53 - Multivariable Calculus | Engineering 183A - Innovation Entrepreneurship | Bioengineering 26

**Phillips Exeter Academy** | High School Diploma Exeter, NH

- 1600 SAT | **Coursework:** Multivariable Calculus | Linear Algebra | Accel. Bio/Chem/Physics *September 2021 - June 2025*

## SKILLS

**Programming:** Python, Java, SQL, MATLAB, TypeScript/JavaScript (Next.js, React), AI/ML (learning)  
**Engineering Tools:** ANSYS, Fusion 360, Onshape, FreeCAD, Arduino, Raspberry Pi, KiCad, 3D Printing  
**Systems/Concepts:** System Performance Optimization, Embedded Systems, Failure Mode Analysis, Python Scripting, Mechatronics  
**Databases/Platforms:** Excel, Word, Powerpoint, GitHub, Redshift (Learning)  
**Soft Skills:** Human-Centered Design, Cross-Functional Collaboration, Stakeholder Engagement, Public Speaking, Mandarin

## WORK/ENGINEERING EXPERIENCE

**CalSol (UC Berkeley Solar Racing Team)** Berkeley, CA  
*Solar Engineer* *September 2025 - Present*

- Designed proj. solar array and bypass diode layout for Gen-11 car, increasing **PV efficiency by 12%** under partial shading.
- Engineered integration of wiring and lightening holes into carbon fiber shell and improved maintenance access

**MUREX Exeter (MATE Marine Robotics Competition Team)** Exeter, NH  
*Mechanical Lead* *November 2021 - June 2024*

- Led design and fabrication of custom 6-DOF underwater ROV and 7-DOF manipulator arm, + independent buoyancy module
- Cut manufacturing costs by **88%** through in-house component design using Onshape + CAM workflows
- Team broke **10 industry records**, including creating the world's **smallest and most affordable network switch**
- Team placed **6th** globally at MATE 2024 and earned awards for Best Engineering Presentation, Poster, and Perfect CAD

**Research: Microplastic Pollution in Rivers (Catholic University of America, CEE Dept)** Washington DC  
*First Author, Researcher* *June 2022 - August 2024*

- Simulated vertical transport of microplastics using flume **hydraulics**; generated **250+** data points used in dispersion modeling
- Designed drone-based sampling protocol to access **3+ previously unreachable river zones**
- **First-author** [research paper](#) selected for WaterSciCon 2024; project awarded by NASA, EPA, GENIUS Olympiad, and ISEF

**Formula 1 in Schools** Exeter, NH  
*Mechanical Lead* *September 2022 - June 2023*

- Designed F1 car using Fusion 360/ANSYS simulations; optimized aerodynamics via iterative testing in wind tunnels
- Achieved **1st place at State Finals 2023** through performance optimization and weight reduction

## TECHNICAL PROJECTS

**Low-Cost Modular Biodiversity Sensing Module (in progress)** | *September 2025 -* Berkeley, CA

- Developing solar-powered wildlife sensing nodes with camera, acoustic, and thermal inputs; lead on **embedded hardware**
- Implementing **Wi-Fi HaLow telemetry** to enable remote, solar-powered sync; prototyping **embedded board with edge AI**
- Collaborating cross-functionally on **AI/ML species ID pipeline** for autonomous biodiversity classification

**Fully Automated Aquaponics System** | *December 2023 - June 2025* Exeter, NH

- Designed and built closed-loop aquaponics system with **automated pH, ammonia, and temperature sensing** using Onshape
- Engineered **integrated biofiltration and nutrient cycling** using siphon effect to optimize water quality and plant growth
- Acquired **iterative feedback** from national/international experts (UNH, Victory Aquaponics, Island School (Bahamas)); led cross-functional ops with faculty/staff to deploy system as live food + edtech solution

## LEADERSHIP

**Innovation Towards the Future (NGO)** Potomac, MD  
*Founder & President* *September 2020 - June 2025*

- Founded global youth NGO **admitted to the UNFCCC** (UN Framework Convention on Climate Change)
- Forged **3 corporate partnerships** to distribute eco-friendly bags/cups to communities in need
- Launched and promoted **global No Single-Use Plastic Campaign** through national and international network connections

**The 28th United Nations Climate Change Conference (COP28) + 18th Conference of Youth (COY18)** Dubai, UAE  
*US Delegate & Global Youth Advocate* *November 2023 - December 2023*

- Led my NGO to **represent the USA** at the **largest global climate conference**, chosen from 5k applicant NGOs
- Participated in formal global climate negotiations, policy formulation, panel discussions, and cultural exhibitions
- Championed youth-driven tech solutions to meet 2030 sustainability targets (renewables, efficiency, divestment)

**National Gallery for America's Young Inventors** Akron, OH  
*Student Board of Directors* *September 2022 - June 2025*

- Evaluated 75+ innovation submissions and selected top six inductees based on impact, feasibility, and market potential